Software Requirements Specification

For

<AI-Integrated Business Automation System (BAS) >

Version 1.0

Prepared by Group #

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Date: January 2026

Team

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1. Introduction:

Organizations must handle intricate company operations across several areas, including finance, human resource management, inventory control, and sales, in the contemporary business environment. Conventional business management systems frequently use manual processes or disconnected software programs, which lead to inefficiencies, decision-making delays, and higher operating expenses.

An intelligent business solution designed for the corporate setting is the AI-Integrated Business Automation System (BAS). This system automates repetitive business tasks by integrating AI and ML technology. It is thought that the Business Automation System (BAS) would assist companies in achieving better business outcomes by integrating predictive analytics and intelligent decision-making. The goal of the Business Automation System (BAS) is to provide a centralized business solution that will enhance company decision-making and expansion in addition to efficiently managing regular business operations.

1.1 Purpose:

Enabling an intelligent, scalable, and automated business solution for controlling and optimizing corporate processes is the goal of the AI-Integrated Business Automation System (BAS). In order to facilitate automation, resource prediction, and real-time business insights, the suggested system would make use of artificial intelligence (AI) and machine learning (ML). By combining finance, HR, inventory, and sales into an intelligent platform, the suggested solution would assist in overcoming the drawbacks of conventional business management solutions.

1.2 Scope:

Creating a cloud-based business management system for businesses is part of the planned Business Automation System's (BAS) scope. The suggested approach would use artificial intelligence (AI) algorithms to automate manual procedures inside a company. The suggested method could be used to manage a number of organizational areas, such as sales, inventories, finance, and human resources. Small and medium-sized businesses (SMEs) searching for digital transformation solutions will benefit from the suggested solution.

1.3 Definitions, abbreviations, and acronyms:

- **BAS** stands for Business Automation System. A system that is interconnected for managing business activities.
- AI, or Artificial intelligence a technique that enables machines to exhibit human-like intelligence.
- **ML** stands for machine learning. a system that enables data-driven machine learning.
- **ERP** stands for Enterprise Resource Planning. A business management software system.
- **KPI** stands for Key Performance Indicator. a set of quantifiable criteria for the company's performance.

1.4 Acronyms and Abbreviations:

- **HR** stands for Human Resources.
- **DBMS** stands for Database Management System.
- **UI** stands for User Interface.
- **API** stands for Application Programming Interface.
- **AWS** Amazon Web Services.

2. Project Planning and Management:

2.1 SWOT Analysis

Strengths:

- AI integration for workflow automation and predictive analytics.
- A modular design for scalability.
- For accessibility, cloud-based.
- An intelligent chatbot and user-friendly interface.

Weaknesses:

- AI need training.
- Data is essential to the effectiveness of AI.
- Skilled developers are needed for complex structures.

Possibilities:

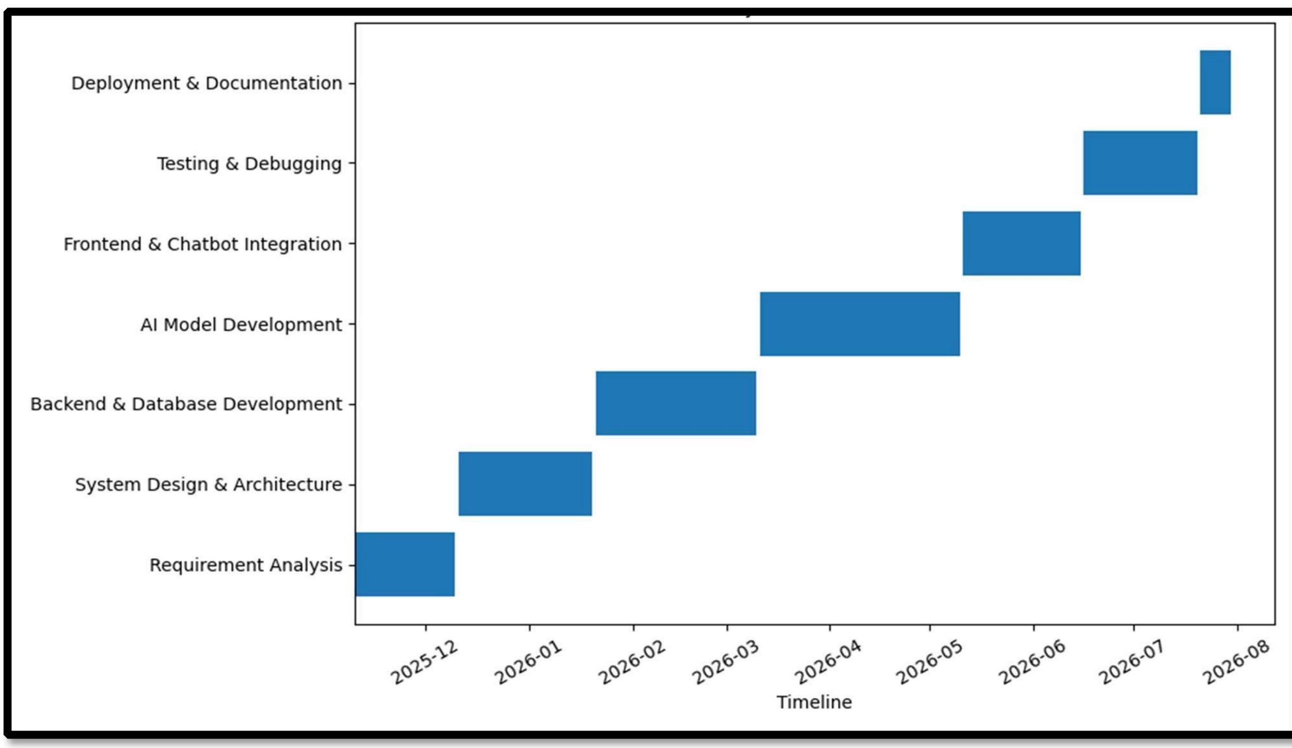
- The need for smart business solutions for SMEs is growing.
- Potential for future blockchain and IoT integration.
- Opportunities to expand cloud-based services for businesses.

Dangers:

- The competition from well-known ERP companies like Oracle and SAP.
- Data security.
- Businesses that use antiquated technologies have inertia.

2.2 Gantt Chart:

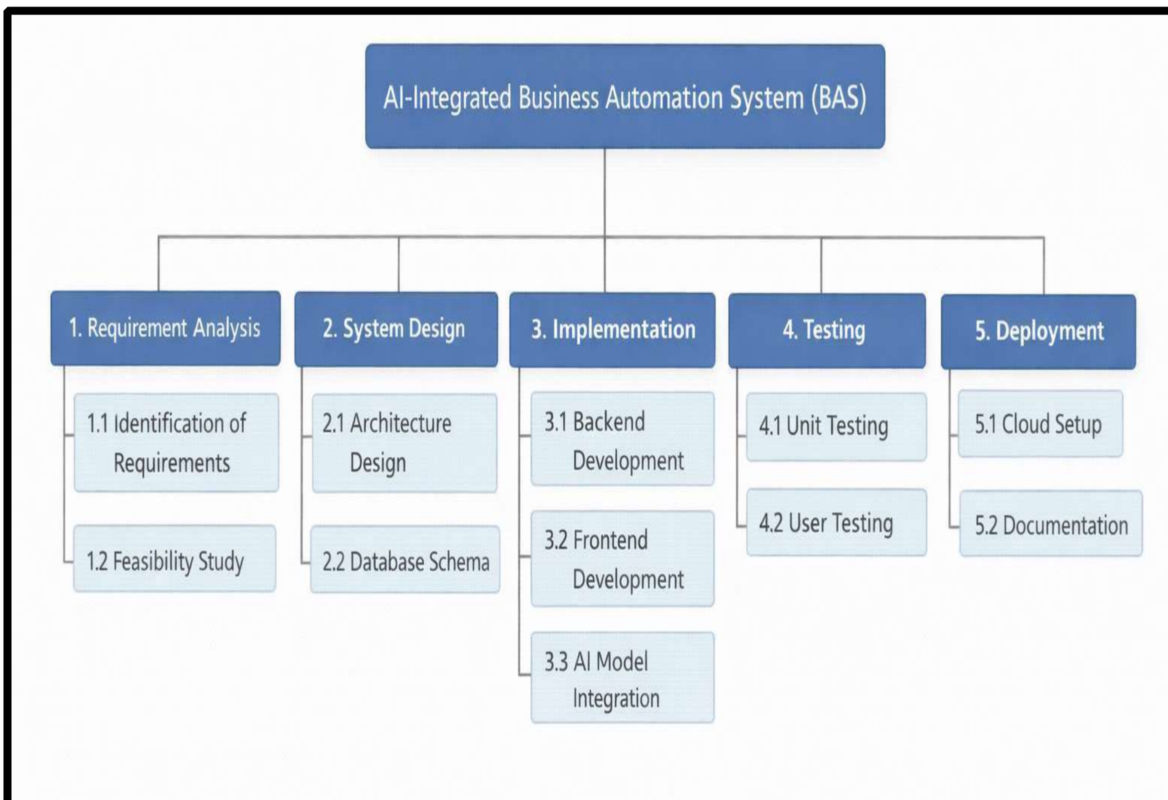
Task	Start Date	End Date	Duration (Days)
Requirement Analysis	Nov10, 2026	Dec10, 2025	30
System Design & Architecture	Dec 11, 2025	Jan 20, 2026	41
Backend & Database Development	Jan 21, 2026	Mar 10, 2026	49
AI Model Development	Mar 11, 2026	May 10, 2026	61
Frontend & Chatbot Integration	May 11, 2026	Jun 15, 2026	36
Testing & Debugging	Jun16, 2026	July 20, 2026	35
Deployment & Documentation	July 21, 2026	July 21, 2026	10



2.3 Work Breakdown Structure (WBS)

- Compiling Requirements
- Design of the System
- Database and Backend Development
- Integration of AI/ML Engines
- Web and Android Frontend Development
- Testing and Integration of Modules
- Maintenance and Deployment

3. Overall Description:



3.1 Product Perspective:

The BAS is an intelligent enterprise management system that offers predictive analytics and automates manual procedures. The BAS system incorporates an AI engine that continuously learns from company data and makes recommendations for business decisions, which sets it apart from conventional ERP systems. The system is cloud-hosted, scalable, and modular in nature. The system integrates with other business modules with ease.

3.2 Product Function:

- **User Authentication:** Dashboards and a safe login system are provided by the system.
- **Finance Module**: The system's offers predictive budgeting and automated spending tracking.
- **HR module:** offers performance analysis, attendance tracking, and automatic hiring.
- **Inventory Module:** AI-based inventory prediction and optimization are provided by the system.
- **Sales Module:** The system offers customer tracking, lead management, and sales forecasting.
- **AI Engine:** The system offers a recommendation system, anomaly detection, and predictive analytics.
- **Chatbot Module:** The system offers a chatbot for simple support and navigation.
- **Dashboard:** Real-time data is shown on a dashboard provided by the system.

3.3 Operating Environment:

- **Front-end:** React.js
- **Backend:** Express Framework (Node.js)
- **Database:** MySQL/PostgreSQL
- **AI Tools:** Python (TensorFlow, Scikit-learn)
- **Chatbot Framework:** Rasa/Dialogflow
- **Deployment:** AWS/Azure (Docker & Kubernetes)
- **Supported Browsers:** Chrome, Firefox, Edge, and Safari.

3.4 Design and Implementation Constraints:

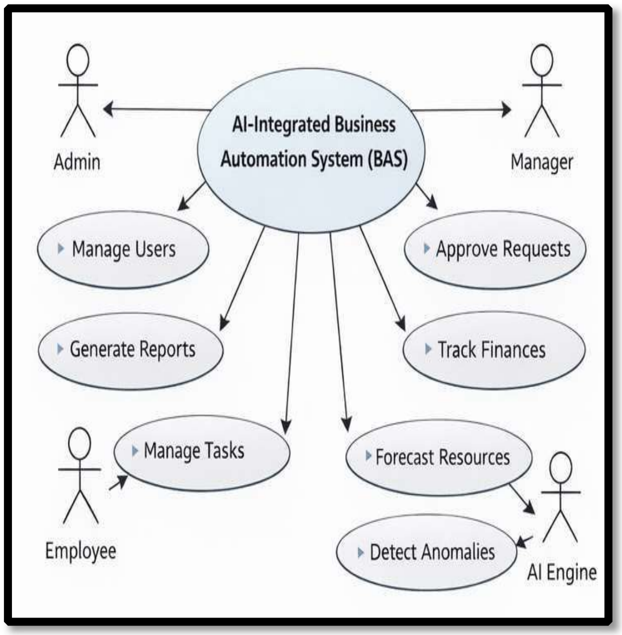
- Time restrictions for resource allocation and project completion.
- Scalability is achieved by using cloud services.
- The intricacy of connecting business modules with the AI engine.
- Ensuring adherence to data privacy rules is crucial.

3.5 Assumptions and Dependencies

- A reliable connection to the internet or server.
- The availability of clean, organized data for the creation of AI models.
- Users should be familiar with fundamental technology, including corporate applications.
- Cloud services are operational.

4. Requirement Identifying Technique:

4.1 Use Case Diagram:



The key players in the AI-Integrated Business Automation System are depicted in the above figure. Different actors engage with the system to carry out different tasks. Administrators, managers, employees, and AI engines are the different actors. The aforementioned system can be used for a number of purposes, including system management, report generation, financial management, resource forecasting, and more. The interactions between the different actors and their corresponding use cases are shown by the arrows.

4.2 Use Case Description:

Use Case Name: Manage Tasks

Analyze This Situation:
Through the BAS interface, employees may effectively examine, add, or change the tasks they have been allocated.
All task-related activities are recorded thanks to the system's real-time synchronization between modules.

Primary Actor:

Employee Precondition:

- The employee has the right credentials and is logged into the BAS system.
- The department or project that the person is assigned to is active in the database.

Manage Tasks (UC1) Main Flow:

- From the dashboard, the employee chooses the "Manage Tasks" use case.
- The current allocated tasks and their statuses are shown by the BAS.
- The employee can mark a job as finished, add new tasks, or change current tasks.
- The BAS keeps track of the task list in real time and sends out deadline alerts.
- The Manager can access the revised task list that has been stored in the database.

UC2-Review Reports Alternative Flow:

- The employee selects "Review Reports" to see the financial or performance data.
- The analytical results produced by the AI Engine are retrieved and displayed by the BAS system.
- The worker examines the data and, if necessary, requests clarification.
- The report can be exported from the system in the preferred forms (PDF, Excel).

Postconditions:

- The employee's modifications to the tasks are successfully updated and recorded by the system.
- The managers can view the updates on their dashboard.
- The AI Engine utilizes the data to forecast and assess performance in the future.

Words Used:

- The term "employee" (E) refers to a person who uses BAS on a daily basis.
- The term "Manager" (M) refers to a person who is in charge of managing the BAS staff.
- AI Engine (AI): The intelligent component of the system that handles data processing uses AI.
- BAS (S): The complete AI-Integrated Business Automation System under development uses BAS.

5. Non-Functional Requirements:

The AI-Integrated Business Automation System's (BAS) non-functional requirements specify the quality characteristics or limitations that guarantee the system's effective, safe, and dependable operation.

5.1 Performance Requirements:

Real-time analytics, visual reporting, and automated replies should all be possible with the least amount of latency. Multiple enterprise users should be able to use it concurrently without experiencing any discernible performance decrease. Under typical operational settings, the dashboard should load in two to three seconds, even when there are big datasets for anomalies or predictive analytics. In order to facilitate fast decision-making across modules like finance, HR, and inventory, the AI models should execute forecasting and data retrieval tasks with the least amount of latency.

5.2 Safety Requirements:

Strict security measures must be implemented in the system to safeguard business-critical data, including financial transactions, personnel information, and inventory records. In the event that hardware or software fails, the system ought to have a backup and recovery mechanism. To avoid data loss or harm to operational data, the system should incorporate validation and error-handling methods. Errors or strange events should be automatically recorded by the system.

5.3 Security Requirements:

To ensure that only authorized users can access data or functions according to their designated roles, the BAS must implement role-based access control. Strong encryption techniques like AES-256 or HTTPS/TLS protocols must be used to encrypt all data both in transit and at rest. Secure login credentials must be used while implementing the user authentication process. For administrator users, two-factor authentication must be used. Common attacks like SQL injection, cross-site scripting, and data exposure must be prevented from affecting the APIs.

5.4 Software Quality Attributes:

All of the system's modules should have an intuitive, user-friendly interface.

High dependability, or precise data processing with high uptime, is what the system should have. The system should be scalable so that it can be readily expanded to accommodate an increase in users, transactions, and data as organizations grow. In order to facilitate user access across many platforms, including desktop, tablet, and mobile, the system's interface must be responsive.

5.5 Business Rules:

Data privacy and business regulations pertaining to employee and corporate financial information management must be followed by the BAS. Payroll computation and expenditure report generation are examples of automated business procedures that must adhere to corporate policy or legal accounting standards. For transparency, the system needs to keep a thorough audit record of all important company operations. AI-based decision-making must be transparent and include explanations for any predictions or suggestions made.

5.6 Interoperability:

Through APIs, the BAS should be able to communicate with current corporate systems like accounting, HRMS, and CRM. For system-to-system communication, it should be able to support various formats like JSON and XML. For enhanced analytics or retraining, the AI Engine ought to be able to communicate with other sources. Future integration with cloud-based services and IoT devices ought to be feasible with little modification.

5.7 Extensibility:

Extensibility should be possible in the system architecture. In order to add or remove features without affecting other features, it must be modular. New business modules, such project management or procurement, should be able to be added with little modification to the code. Without changing the structure of the system, the AI Engine design should enable the inclusion of new machine learning algorithms or model upgrades. Changes to intents and training data should not impact service availability thanks to the nature of the chatbot platform.

5.8 Maintainability:

To make troubleshooting, version control, and future updates easier, every system component needs to be thoroughly documented. To make maintenance and troubleshooting easier, best practices like clean, modular, and reusable code must be adhered to. To enable future upgrades, configuration files, system deployment scripts, and AI model parameters must all be kept in a repository. Log monitoring, software patching, and model retraining are all necessary components of maintenance programs.

5.9 Portability:

Platform independence refers to the BAS's ability to function on a variety of operating systems, including Windows, Linux, and macOS. It should function reliably across a variety of browsers, including Microsoft Edge, Mozilla Firefox, Google Chrome, and Safari. The program will be executed in several contexts, including testing, staging, and production, and it should be able to run reliably utilizing containerization technologies like Docker. It should work with platforms like AWS, Azure, Google Cloud, and others for cloud deployment.

5.10 Reusability:

To enable quicker system development in the future, the system must be built with reusable components including data visualization, authentication, and API integration. To enable reuse in other enterprise systems, the AI models, system dashboard, or automation workflow must be structured as microservices. To make integration with next projects easier, the system must have standardized interfaces that are adequately documented. To make reuse easier, utility functions must be abstracted.

5.11 Installation:

The BAS should come with thorough instructions and a straightforward installation and configuration procedure. One-click installation via automated scripts like Docker or CI/CD pipelines should be an option in the installation process. To make it easier to change environment variables like database URLs, API keys, and cloud credentials, configuration files should be provided. Additionally, automatic prerequisite checks should be incorporated into the installation procedure to avoid future compatibility issues. Additionally, automated patching tools should be used to perform updates on a regular basis.

6.Other Requirements:

6.1 Online Documentation: FAQs and an interactive user handbook.

6.2 Purchased Requirements: AWS Cloud and Rasa licenses.

6.3 Licensing Requirements: MIT-licensed open-source software is required.

6.4 Legal & Copyright Notices: Adherence to regional data protection regulations.

6.5 Applicable Standards: EE 830-1998, ISO 9001, and GDPR.

6.6 Reports:

- Reports on Financial Summaries
- Reports on Sales Forecasts
- Reports on Inventory and HR Analytics
- Dashboard for Executive Summary

7.References:

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- Rasa Framework Documentation – https://rasa.com/docs/
- AWS Developer Guide – https://docs.aws.amazon.com/